

Hypertension

- **Target organ damage:** Damage to vital organs like the heart, kidneys, brain, and eyes caused by prolonged high blood pressure.
 - **Pathogenesis:** The development or progression of a disease.
 - **Multifactorial:** Involving many factors, such as genetic, lifestyle, and environmental.
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Secondary Hypertension

- **Renovascular stenosis:** Narrowing of arteries that supply blood to the kidneys.
 - **Glomerulonephritis:** Inflammation of the kidney's filtering units (glomeruli).
 - **Polycystic kidney disease:** A genetic condition causing fluid-filled cysts in the kidneys.
 - **Pheochromocytoma:** A rare tumor of the adrenal glands that produces excessive adrenaline.
 - **Cushing's syndrome:** A hormonal disorder caused by high levels of cortisol.
 - **Hyperaldosteronism:** Overproduction of aldosterone, a hormone regulating salt and water balance.
 - **Acromegaly:** A disorder caused by excessive growth hormone leading to enlarged body parts.
 - **Coarctation of the aorta:** Narrowing of the major artery (aorta), restricting blood flow.
 - **Obstructive sleep apnoea:** A condition where breathing stops and starts repeatedly during sleep.
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Classification of Blood Pressure in Adults

- **Prehypertension:** Blood pressure that is above normal but not yet in the hypertensive range.
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Clinical Features

- **Occipital headache:** Pain in the back of the head or upper neck.
- **Epistaxis:** Nosebleeds.
- **Hematuria:** Blood in the urine.
- **TIA (Transient Ischemic Attack):** A temporary reduction in blood supply to the brain, often called a "mini-stroke."

- **Thyrotoxicosis:** Excessive thyroid hormone levels in the blood.
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Physical Examination Findings

- **Truncal obesity:** Excess fat concentrated around the abdomen and chest.
 - **Radiofemoral delay:** A delay in the pulse between the radial (wrist) and femoral (groin) arteries, indicating arterial blockage.
 - **Chronic pyelonephritis:** Long-term kidney infection causing scarring and recurrent pain.
 - **Abdominal bruit:** Abnormal sound heard during a physical examination, indicating narrowed blood vessels.
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Risk Factors

- **Hyperlipidaemia:** High levels of fats (cholesterol and triglycerides) in the blood.
- **CAD (Coronary Artery Disease):** A condition where blood vessels supplying the heart are narrowed or blocked.

Complications of Hypertension

1. **CNS Complications**
 - **CVA (Cerebrovascular Accident):** A stroke caused by interrupted blood supply to the brain.
 - **Subarachnoid hemorrhage:** Bleeding into the space surrounding the brain.
 - **Hypertensive encephalopathy:** Brain dysfunction due to extremely high blood pressure, often reversible.
 - **Papilloedema:** Swelling of the optic disc due to increased intracranial pressure.
2. **Ophthalmic Complications**
 - **Hypertensive retinopathy:** Damage to the retina caused by high blood pressure.
 - **Arteriovenous nipping:** Compression of veins by thickened arterioles in the retina.
 - **Exudates:** Fluid leakage from blood vessels into surrounding tissues.
3. **Cardiovascular Complications**
 - **Myocardial infarction:** Heart attack due to blocked blood flow.
 - **Aortic aneurysm:** Abnormal bulging of the aorta.
 - **Aortic dissection:** Tearing of the aortic wall layers.
4. **Renal Complications**
 - **Proteinuria:** Presence of excess protein in the urine.

- **Progressive renal failure:** Gradual decline in kidney function.
 - 5. **Malignant Hypertension**
 - **Markedly high BP:** Exceptionally elevated blood pressure causing severe complications.
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Investigations in Hypertension

1. **Basic Investigations for All Patients**
 - **Blood urea nitrogen:** A test to evaluate kidney function.
 - **Serum creatinine:** A marker for kidney function.
 - **Serum electrolytes:** Measures levels of sodium and potassium in the blood.
 - **Serum lipid profile:** Tests cholesterol and triglyceride levels.
 - **ECG (Electrocardiogram):** Measures electrical activity of the heart.
 2. **Special Group of Patients**
 - **Renal ultrasonography:** Imaging test to examine kidney structure.
 - **Angiography:** Imaging of blood vessels.
 - **TSH (Thyroid-stimulating hormone):** Hormone test for thyroid function.
 - **Urinary cortisol:** Test for cortisol levels in urine, indicating adrenal gland function.
 - **Plasma renin activity:** Measures renin enzyme levels related to blood pressure regulation.
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Treatment of Hypertension

1. **General Measures (Lifestyle Modifications)**
 - **Biofeedback:** A technique to control bodily functions like heart rate using feedback from monitoring devices.
 - **Salt restriction:** Limiting sodium intake to reduce blood pressure.
 - **Weight reduction:** Losing weight to improve overall health and lower BP.

Antihypertensive Drug Therapy

A. Diuretics

- **Thiazide diuretics:** Medications that increase urine output by acting on the kidney's distal tubules.
- **Loop diuretics:** Drugs that act on the loop of Henle in the kidney to remove excess fluid.

- **Potassium-sparing diuretics:** Medications that help retain potassium while increasing urine output.
 - **Aldosterone receptor blockers:** Medications that inhibit aldosterone to prevent sodium retention and potassium loss.
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B. Beta-blockers

- **Cardioselective beta-blockers:** Drugs that primarily block beta-1 receptors in the heart, reducing heart rate and blood pressure.
 - **Heart block:** A condition where the heart's electrical signals are slowed or blocked.
 - **COPD (Chronic Obstructive Pulmonary Disease):** A lung disease causing breathing difficulties.
 - **Peripheral vascular disease:** Poor blood circulation in the limbs.
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C. Combined Alpha- and Beta-blockers

- **Carvedilol/Labetalol:** Medications that block both alpha and beta receptors, reducing blood pressure.
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D. ACE Inhibitors

- **ACE inhibitors (Angiotensin-Converting Enzyme inhibitors):** Drugs that lower blood pressure by relaxing blood vessels and reducing fluid retention.
 - **Angioedema:** Swelling under the skin, often due to a drug reaction.
 - **Bilateral renal artery stenosis:** Narrowing of both arteries supplying blood to the kidneys.
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E. Angiotensin II Antagonists (ARBs)

- **Angiotensin II:** A hormone that narrows blood vessels and increases blood pressure.
 - **ARBs (Angiotensin Receptor Blockers):** Medications that block angiotensin II effects, reducing blood pressure.
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F. Calcium Channel Blockers

- **Arterial vasodilatation:** Widening of arteries to reduce blood pressure.
 - **Raynaud's disease:** A condition causing reduced blood flow to fingers and toes, often triggered by cold or stress.
 - **Pulmonary hypertension:** High blood pressure in the arteries of the lungs.
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G. Alpha Blockers

- **Prazosin/Terazosin/Doxazosin:** Medications that relax blood vessels by blocking alpha receptors, lowering blood pressure.
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H. Direct Vasodilators

- **Vasodilatation:** Relaxation of blood vessel walls, increasing their diameter to lower blood pressure.
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I. Central α_2 -blockers and Other Central Agents

- **Central α_2 -blockers:** Medications that act on the central nervous system to reduce blood pressure by decreasing nerve signals.
- **Clonidine/Alpha-methyldopa/Reserpine:** Drugs that lower blood pressure by calming the nervous system.

Other Drug Groups

- **Alpha adrenergic receptor blockers:** Medications that relax blood vessels by blocking alpha receptors, reducing blood pressure.
 - **Centrally acting agents:** Drugs that act on the brain to lower blood pressure by reducing nerve signals.
 - **Vasodilators:** Medications that relax and widen blood vessels, improving blood flow.
 - **Mixed alpha- and beta-blockers:** Drugs that block both alpha and beta receptors to lower blood pressure.
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Broad Guidelines for Drug Selection

- **Ischemic heart disease:** A condition where reduced blood flow to the heart causes chest pain or heart attacks.
 - **Peripheral vascular disease:** A disorder of blood circulation in the arms or legs.
 - **Renovascular hypertension:** High blood pressure caused by narrowing of arteries supplying the kidneys.
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Guidelines for Treatment Based on Hypertension Stage

- **Prehypertension:** Slightly elevated blood pressure not yet in the hypertensive range.
- **Stage 1 Hypertension:** Moderate elevation in blood pressure requiring single-drug treatment.
- **Stage 2 Hypertension:** Significant elevation in blood pressure often requiring a combination of medications.

Ischemic heart disease

Definition

- **Ischemic heart disease (IHD):** A condition caused by reduced blood supply to the heart muscles, leading to oxygen deficiency.
 - **Atherosclerosis:** Fatty deposits that build up inside arteries, causing narrowing or blockage.
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Causes

- **Thrombosis:** Formation of a blood clot within a blood vessel, blocking blood flow.
 - **Coronary artery spasm:** Sudden, temporary narrowing of coronary arteries due to muscle contraction.
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Risk Factors

- **Hyperlipidemia:** Elevated levels of cholesterol or fats in the blood.
 - **Sedentary lifestyle:** Lack of regular physical activity.
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Symptoms

- **Angina pectoris:** Chest pain or discomfort due to insufficient blood supply to the heart.
 - **Palpitations:** Feeling of a fast or irregular heartbeat.
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Diagnosis

- **ECG (Electrocardiogram):** A test recording the electrical activity of the heart.
 - **Echocardiogram:** An ultrasound to check heart structure and function.
 - **Coronary angiography:** Imaging test to visualize coronary artery blockages.
 - **Cardiac markers (e.g., troponin):** Proteins released into the blood after heart muscle damage.
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Treatment

- **Antiplatelets:** Drugs like aspirin that prevent blood clots.
 - **Nitrates:** Medications that dilate blood vessels and relieve chest pain.
 - **Statins:** Medications to lower cholesterol and reduce plaque buildup in arteries.
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Surgical Treatment

- **Angioplasty:** A procedure using a balloon to widen narrowed arteries.
 - **Stent:** A tiny mesh tube used to keep arteries open.
 - **Coronary artery bypass grafting (CABG):** A surgery that creates a new pathway for blood around blocked arteries.
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Prevention

- **Stress management:** Techniques to reduce mental or emotional strain.
- **Regular check-ups:** Routine medical evaluations to detect risk factors early.

Cushing syndrome

Causes

- **Cortisol:** A hormone produced by the adrenal glands that helps regulate metabolism, immune response, and stress.

- **Pituitary adenoma:** A noncancerous tumor in the pituitary gland causing excess hormone production.
 - **Ectopic ACTH secretion:** Production of ACTH by non-pituitary tumors, often in the lungs.
 - **Adrenal tumors:** Abnormal growth in adrenal glands causing excessive hormone secretion.
 - **Exogenous corticosteroid use:** Taking external steroids for long periods, leading to hormone imbalance.
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Clinical Features

- **Moon face:** A rounded, puffy face due to fat accumulation.
 - **Buffalo hump:** Fat buildup on the upper back.
 - **Striae:** Streak-like marks on the skin, usually purple in Cushing syndrome.
 - **Hyperglycemia:** High blood sugar levels.
 - **Hirsutism:** Unusual hair growth in women, often on the face and body.
 - **Amenorrhea:** Absence of menstrual periods.
 - **Oligomenorrhea:** Infrequent menstrual cycles.
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Diagnosis

- **24-hour urinary cortisol excretion:** Measures cortisol levels in the urine collected over a day.
 - **Dexamethasone suppression test:** A test to see if synthetic steroids can suppress cortisol production.
 - **ACTH:** A hormone produced by the pituitary gland that stimulates cortisol production in adrenal glands.
 - **CT scan/MRI:** Imaging techniques to detect tumors.
 - **Late-night salivary cortisol:** A test measuring cortisol levels in saliva, typically low at night in healthy individuals.
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Treatment

- **Pituitary surgery:** Surgical removal of tumors from the pituitary gland.
- **Adrenalectomy:** Surgical removal of one or both adrenal glands.
- **Ketoconazole:** A drug that inhibits cortisol production.
- **Metyrapone:** A medication that blocks cortisol synthesis.

- **Mitotane:** A drug used to decrease adrenal gland activity, particularly in adrenal tumors.
- **Cabergoline:** A medication that reduces ACTH secretion in some pituitary-related cases.
- **Radiation therapy:** Use of high-energy rays to destroy abnormal tissue.

Mitral stenosis

Causes

- **Rheumatic heart disease:** A chronic heart condition caused by untreated or poorly treated streptococcal infection.
 - **Calcification:** The accumulation of calcium deposits in tissues, leading to stiffness and narrowing of the mitral valve.
 - **Congenital:** A condition present from birth.
 - **Degenerative:** Related to gradual deterioration due to aging.
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Pathophysiology

- **Orifice:** The opening of the mitral valve through which blood flows.
 - **Fibrosis:** Thickening or scarring of tissue, causing reduced elasticity.
 - **Commissures:** The points where the leaflets of the valve meet.
 - **Chordae tendineae:** Tendon-like cords connecting the valve leaflets to the heart muscle.
 - **Hemodynamically significant:** Having a critical effect on blood flow.
 - **Pulmonary venous congestion:** Blood backing up in the pulmonary veins due to increased pressure.
 - **Pulmonary hypertension:** High blood pressure in the lungs' blood vessels.
 - **Atrial fibrillation (AF):** An irregular and often rapid heart rhythm.
 - **Thrombus:** A blood clot within a blood vessel or the heart.
 - **Thromboembolism:** A condition where a blood clot travels to block another vessel.
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Clinical Manifestations

- **Orthopnoea:** Difficulty breathing while lying flat.
- **Paroxysmal nocturnal dyspnoea (PND):** Sudden shortness of breath during sleep.
- **Acute pulmonary oedema:** Sudden fluid accumulation in the lungs causing difficulty breathing.
- **Hemoptysis:** Coughing up blood.

- **Thromboembolic complications:** Blockage of blood flow to vital organs, leading to strokes or limb ischemia.
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Signs

- **Mitral facies:** Flushed cheeks with pale skin elsewhere, indicative of chronic mitral stenosis.
 - **Auscultation:** Listening to the heart sounds using a stethoscope.
 - **Right ventricular heave:** A palpable pulsation of the right side of the heart due to strain.
 - **P2:** The second heart sound caused by pulmonary valve closure.
 - **Crackles:** Crackling sounds heard during lung auscultation, indicating fluid.
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Investigations

- **ECG (Electrocardiogram):** Measures the electrical activity of the heart.
 - **Echocardiography:** Ultrasound imaging to visualize heart structure and function.
 - **Cardiac catheterization:** A procedure using a thin tube to evaluate heart function and blood flow.
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Treatment

A. Medical Management

- **Sodium restriction:** Reducing salt intake to manage fluid retention.
- **Diuretics:** Medications that help eliminate excess fluid from the body.
- **Digoxin:** A medication used to control heart rate in atrial fibrillation.
- **Anticoagulants:** Drugs like warfarin to prevent blood clots.

B. Surgical Management

- **Percutaneous balloon valvotomy:** A minimally invasive procedure to widen the mitral valve using a balloon.
- **Open valvotomy:** A surgical procedure to manually separate the valve leaflets.
- **Mitral valve replacement:** Surgery to replace the diseased valve with an artificial one.

Myocardial Infarction (MI)

Myocardial Infarction (MI)

- **Necrosis:** Death of tissue
- **Thrombus:** Blood clot
- **Plaque rupture:** Break in fatty deposit in arteries

Types

- **Subendocardial:** Inner layer of the heart
- **Transmural:** Full thickness of the heart
- **Silent:** Without noticeable symptoms

Symptoms

- **Syncope:** Fainting
- **Arrhythmias:** Abnormal heart rhythms

Physical Signs

- **Sympathetic:** Body's fight-or-flight response
- **Vagal:** Related to the vagus nerve (rest/digest)
- **Hypotension:** Low blood pressure
- **Oliguria:** Low urine output
- **JVP:** Jugular venous pressure (indicator of heart function)
- **Crackles:** Crackling sounds in the lungs

Complications

- **Cardiogenic shock:** Severe heart failure causing low blood pressure
- **Pulmonary edema:** Fluid in the lungs
- **Papillary muscle infarction:** Death of heart muscle affecting valve function
- **Septal rupture:** Hole in heart's septum (wall between chambers)
- **Cardiac tamponade:** Fluid around the heart restricting function
- **Embolism:** Blockage in blood vessels, can affect brain, lungs, etc.
- **Dressler's syndrome:** Inflammation after heart injury

Investigations

- **Leucocytosis:** High white blood cell count
- **ESR:** Erythrocyte sedimentation rate (marker of inflammation)
- **ECHO:** Ultrasound of the heart

Management

- **Reperfusion:** Restoring blood flow to the heart
- **PCI:** Percutaneous coronary intervention (angioplasty)

Specific Therapy

- **Thrombolysis:** Breaking down blood clots
- **ACE Inhibitors:** Medications that help relax blood vessels
- **Antiplatelets:** Medications that prevent blood clots
- **Anticoagulants:** Medications to prevent blood clotting

Thrombolysis

- **Indications:** Reasons for treatment
- **Reperfusion:** Restoring blood flow to tissue
- **Complications:** Unwanted effects

PCI (Percutaneous Intervention)

- **Glycoprotein IIb/IIIa inhibitors:** Drugs preventing clot formation

Right Ventricular MI

- **Hypotension:** Low blood pressure
- **Diuretics:** Medications that remove excess fluid from the body

Rehabilitation

- **Mobilization:** Gradual movement to regain strength
- **Hypokalemia:** Low potassium levels in the blood

Heart Failure (Cardiac Failure)

Pathophysiology

- **Preload:** The initial stretching of the heart muscle (ventricles) before contraction, caused by blood returning to the heart.
- **Afterload:** The pressure the heart must work against to eject blood.
- **Myocardial contractility:** The ability of the heart muscle to contract and pump blood, influenced by nervous system signals and hormones.

Heart Failure Characteristics

- **Cardiac output:** The amount of blood the heart pumps in a minute.

- **Renin-angiotensin system:** A hormone system that regulates blood pressure and fluid balance.
- **Autonomic nervous system:** The part of the nervous system that controls involuntary functions like heart rate and blood pressure.

Compensatory Mechanisms

- **Frank-Starling law:** Describes the relationship between the heart's filling and its ability to pump blood; the more the heart fills, the more it pumps, to a point.
- **Ventricular compliance:** The ability of the heart's ventricles to stretch and accommodate blood without losing function.
- **Sympathetic stimulation:** Activation of the part of the nervous system that increases heart rate and blood pressure in response to stress.

Myocardial Remodelling

- **Hypertrophy:** The enlargement of heart muscle cells.
- **Apoptosis:** Programmed cell death.
- **Extracellular matrix:** The network of proteins and other substances that provide structural support to cells.

Types of Heart Failure

- **Acute Heart Failure:** Sudden onset failure, leading to inadequate blood supply.
- **Chronic Heart Failure:** Gradual onset, often with long-term symptoms.
- **Compensated Heart Failure:** When the body's compensatory mechanisms prevent overt failure symptoms.
- **Forward Heart Failure:** Decreased blood flow to the body despite normal output.
- **Backward Heart Failure:** Blood accumulates in the venous system due to salt and water retention.

Aetiology (Causes of Heart Failure)

- **Systemic hypertension:** High blood pressure throughout the body.
- **Pulmonary hypertension:** High blood pressure in the lungs.
- **Aortic stenosis:** Narrowing of the aortic valve, restricting blood flow.
- **Mitral regurgitation:** Backflow of blood from the left ventricle into the left atrium.
- **Ventricular septal defect:** A hole in the wall separating the heart's ventricles.
- **Myocarditis:** Inflammation of the heart muscle, often due to infection.
- **Cardiomyopathy:** Disease of the heart muscle.

Clinical Manifestations

- **Dyspnoea:** Difficulty breathing.
- **Orthopnoea:** Difficulty breathing when lying flat.
- **Paroxysmal Nocturnal Dyspnoea (PND):** Sudden shortness of breath at night.
- **Cardiac Asthma:** Wheezing and coughing caused by heart failure.
- **Cheyne-Stokes Respiration:** Abnormal breathing pattern alternating between deep breaths and periods of no breathing.
- **Oliguria:** Low urine output.
- **Cardiac Oedema:** Swelling due to fluid accumulation, typically in the lower legs.
- **Cyanosis:** Bluish discoloration of the skin due to lack of oxygen.
- **Pulsus alternans:** Alternating strong and weak pulses.
- **Hypotension:** Low blood pressure.

Diagnostic Studies

- **Cardiomegaly:** Enlarged heart, visible on X-ray.
- **Kerley lines:** Radiographic lines in the lungs indicating fluid accumulation.
- **Ejection fraction:** The percentage of blood the heart pumps out with each contraction.
- **BNP Levels:** B-type natriuretic peptide, elevated in heart failure.

Management of Heart Failure

- **Diuretics:** Medications to remove excess fluid.
- **Vasodilators:** Medications that relax and widen blood vessels.
- **Inotropic Agents:** Drugs that increase heart contractility, such as digitalis.
- **Beta Blockers:** Medications that reduce heart rate and blood pressure to reduce workload on the heart.

Acute Rheumatic Fever (ARF)

Pathogenesis

- **Antigen:** A substance that triggers the immune system to produce antibodies.
- **Autoimmune:** When the body's immune system mistakenly attacks its own tissues.

Diagnosis (Jones's Criteria)

- **Polyarthrititis:** Inflammation and pain in multiple joints.
- **Carditis:** Inflammation of the heart layers (pericardium, myocardium, or endocardium).
- **Chorea:** Sudden, involuntary, and irregular movements.
- **Subcutaneous nodules:** Small lumps under the skin.

- **Erythema marginatum:** A skin rash with a clear center and wavy edges.
- **ESR (Erythrocyte Sedimentation Rate):** A blood test indicating inflammation.
- **CRP (C-reactive protein):** A marker of inflammation in the blood.
- **PR interval:** The time between electrical impulses in the heart, measured on an ECG.

Clinical Manifestations

- **Tachycardia:** Rapid heart rate.
- **Cardiomegaly:** Enlarged heart.
- **Sydenham's chorea:** A specific type of chorea associated with ARF.

Laboratory Features

- **Leucocytosis:** Increased white blood cell count.
- **ASO titer (Antistreptolysin O):** A blood test measuring antibodies against streptococcal infection.
- **ECG (Electrocardiogram):** A test recording the electrical activity of the heart.

Management

- **Salicylates:** A class of drugs like aspirin used to reduce pain and inflammation.
- **Corticosteroids:** Anti-inflammatory drugs for severe symptoms.
- **Congestive heart failure:** A condition where the heart is unable to pump blood effectively.
- **Valvular lesions:** Damage or abnormalities in the heart valves.

Prevention

- **Prophylaxis:** Measures to prevent disease, such as continuous antibiotic use.
- **Mass prophylaxis:** Large-scale preventive treatment in populations at risk.

Disseminated Intravascular Coagulation (DIC)

Definition

- **Coagulation:** The process of blood clot formation.
- **Clotting factors:** Proteins in the blood that help it clot.
- **Platelets:** Blood cells that assist in clotting.
- **Organ damage:** Harm to organs due to disrupted blood flow.

Causes

- **Sepsis:** A life-threatening infection causing inflammation.

- **Placental abruption:** Premature separation of the placenta.
- **Eclampsia:** Severe complications of pregnancy marked by seizures.
- **Leukemia:** A type of blood cancer.
- **Hypovolemic shock:** A condition due to significant blood or fluid loss.
- **Transfusion reactions:** Adverse effects of blood transfusion.

Pathophysiology

- **Endothelial injury:** Damage to the inner lining of blood vessels.
- **Coagulation cascade:** A series of steps where clotting factors interact to form a clot.
- **Microthrombi:** Tiny blood clots blocking small blood vessels.
- **Ischemia:** Restricted blood supply to tissues.

Clinical Features

1. **Bleeding**
 - **Petechiae:** Small red or purple spots due to bleeding under the skin.
 - **Purpura:** Larger areas of bleeding under the skin.
 - **Melena:** Black, tarry stools due to gastrointestinal bleeding.
 - **Hematemesis:** Vomiting blood.
2. **Organ Dysfunction**
 - **Renal failure:** Kidneys stop functioning properly.
 - **Liver dysfunction:** Impaired liver function.
3. **Thrombotic Features**
 - **Gangrene:** Tissue death due to lack of blood supply.
 - **Cerebrovascular accidents:** Stroke caused by blood clots in the brain.

Diagnosis

- **PT (Prothrombin Time):** A test to measure blood clotting time.
- **aPTT (Activated Partial Thromboplastin Time):** Measures the time for blood to clot.
- **Fibrinogen:** A protein essential for clot formation.
- **D-dimer:** A substance released when clots break down.
- **Schistocytes:** Fragmented red blood cells seen in blood smears.
- **FDPs (Fibrin Degradation Products):** Substances released when fibrin breaks down.
- **Thrombin time:** The time for fibrin formation in the blood.

Management

1. **Supportive Treatment**
 - **Fresh frozen plasma (FFP):** Contains clotting factors to control bleeding.

- **Cryoprecipitate:** A blood product rich in fibrinogen.
- 2. **Anticoagulation**
 - **Heparin:** A blood thinner to prevent clotting.
- 3. **Organ Support**
 - **Dialysis:** A process to remove waste from the blood during kidney failure.
 - **Ventilator:** A machine to assist breathing during respiratory failure.

Prognosis

- **Survival rates:** The likelihood of recovery depending on timely treatment.

Anticoagulant Drugs, Uses, and Precautions in Dental Practice

1. Warfarin

- **Thrombosis:** Formation of blood clots in blood vessels.
- **Atrial fibrillation:** An irregular and often rapid heart rhythm.
- **Deep vein thrombosis (DVT):** A blood clot in a deep vein, usually in the leg.
- **Pulmonary embolism (PE):** A blockage in one of the pulmonary arteries in the lungs.
- **INR (International Normalized Ratio):** A test to measure blood clotting and monitor warfarin therapy.

2. Heparin

- **Myocardial infarction:** Commonly known as a heart attack, caused by blocked blood flow to the heart.
- **Low molecular weight heparin (e.g., enoxaparin):** A modified form of heparin used for preventing clots.

3. Direct Oral Anticoagulants (DOACs)

- **Apixaban, Rivaroxaban, Dabigatran:** Commonly used modern anticoagulant drugs.
- **DVT (Deep Vein Thrombosis):** A blood clot in a vein, usually in the legs.
- **PE (Pulmonary Embolism):** A blood clot in the lung arteries.
- **Half-life:** The time taken for the concentration of a drug to reduce by half in the body.

4. Aspirin

- **Platelet inhibitor:** A substance that prevents platelets from clumping together, reducing clot formation.

- **Cardiovascular disease:** Disorders of the heart and blood vessels, such as heart attack and stroke.

5. Clopidogrel (Plavix)

- **Stent:** A tube inserted into a blood vessel to keep it open.
- **Platelet aggregation:** The clumping together of platelets in the blood to form clots.

6. Dipyridamole

- **Stroke prevention:** Measures taken to reduce the risk of strokes caused by blood clots.
- **Prosthetic heart valves:** Artificial devices implanted to replace damaged heart valves.

General Precautions in Dental Practice for Anticoagulants

- **Medical history:** A record of past and current health conditions and treatments.
- **Local hemostatic measures:** Techniques or agents to control bleeding, such as gelatin sponges or bone wax.

Signs and Symptoms of Angina and Case Management

Angina

- **Myocardial ischemia:** Reduced blood flow to the heart muscle.
- **Atheroma:** Fatty deposits in the walls of arteries.
- **Coronary arterial spasm:** Sudden narrowing of the coronary arteries.

Worsening Factors

- **Anaemia:** A condition with low red blood cell count or hemoglobin.
 - **Tachycardia:** Abnormally fast heart rate.
 - **Hyperthyroidism:** Overactive thyroid gland producing excess thyroid hormone.
 - **Aortic stenosis:** Narrowing of the aortic valve.
 - **Aortic regurgitation:** Leaking of the aortic valve causing blood to flow back into the heart.
 - **Arrhythmias:** Irregular heartbeats.
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Clinical Features

Pain

- **Retro-sternal:** Behind the sternum (breastbone).
- **Sublingual nitrates:** Medications placed under the tongue to relieve pain.

Types of Angina

1. **Angina decubitus:** Pain occurring while lying down.
 2. **Nocturnal angina:** Pain at night, often associated with aortic regurgitation, nightmares, sweating, and a wide pulse pressure.
 3. **Prinzmetal's angina:** Caused by coronary artery spasm, showing transient ST elevation on ECG.
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Examination

Physical Signs During Pain

- **Fourth heart sound:** A low-frequency sound caused by atrial contraction.
 - **Mitral regurgitation murmur:** Backward blood flow through the mitral valve.
 - **Paradoxical splitting:** Abnormal separation of heart sounds during expiration.
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Investigations

1. **ECG:** Test measuring electrical activity of the heart; shows reversible ST depression/elevation.
 2. **Myocardial perfusion scan:** Imaging test using radioactive thallium to assess blood flow.
 3. **Coronary arteriography:** X-ray imaging to identify narrowing or blockage in coronary arteries.
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Management

General Measures

- **Hyperlipidemia:** High cholesterol levels.
- **Diabetes:** Chronic condition affecting blood sugar levels.
- **Valvular diseases:** Disorders affecting heart valves.

Drug Treatment

1. **Nitrates:**
 - **Glyceryl trinitrate:** Relieves pain within minutes.
 - **Isosorbide dinitrate/mononitrate:** Long-acting nitrates to prevent chest pain, requiring nitrate-free intervals to avoid tolerance.
2. **Beta-blockers:**
 - Reduce heart rate, myocardial oxygen demand, and anxiety.
 - **Examples:** Propranolol, Metoprolol, Atenolol.
3. **Calcium antagonists:**
 - Vasodilators reducing cardiac excitability.
 - **Examples:** Nifedipine, Verapamil, Diltiazem.
4. **Antiplatelet agents:**
 - **Aspirin:** Prevents clot formation by inhibiting thromboxane A2.
 - **Clopidogrel:** Alternative for aspirin-intolerant patients.

Surgical Treatment

- **CABG (Coronary Artery Bypass Grafting):** Surgery using veins or arteries as grafts to bypass blockages.
- **PTCA (Percutaneous Transluminal Coronary Angioplasty):** Balloon procedure, often with drug-coated stents to prevent re-blockage.

Aetiology and Clinical Features of Subacute Bacterial Endocarditis (SBE)

Subacute Bacterial Endocarditis (SBE)

- **Microbial infection:** Infection caused by microorganisms in the heart's inner lining or valves.
- **Endothelium:** The thin layer of cells lining the inside of the heart.
- **Endarteritis:** Infection in arteries, commonly in areas with abnormal blood flow like AV shunts, PDA, or coarctation of the aorta.

Pathology

- **Platelet-fibrin deposition:** Formation of a non-bacterial thrombus due to damaged endothelium.
- **Vegetations:** Clumps of bacteria and platelets that adhere to the heart lining.
- **Viridans streptococci:** Bacteria from the mouth commonly causing subacute infections.
- **Staphylococcus aureus:** Common cause of acute infections.
- **HACEK group:** A group of slow-growing bacteria causing prosthetic valve infections.

Clinical Features

General Symptoms

- **Fever:** Persistent, low-grade temperature.
- **Night sweats:** Excessive sweating during sleep.
- **Weight loss:** Unexplained loss of body weight.

Cardiac Symptoms

- **Heart murmur:** Abnormal heart sound due to turbulent blood flow.
- **Heart failure:** Reduced ability of the heart to pump blood efficiently.
- **Heart blocks:** Delays or interruptions in the heart's electrical system.

Extracardiac Symptoms

- **Clubbing:** Bulging of the fingertips.
- **Splenomegaly:** Enlarged spleen.
- **Petechiae:** Small red or purple spots due to minor bleeding.
- **Janeway lesions:** Painless red spots on palms and soles.
- **Osler nodes:** Painful red nodules on fingers or toes.
- **Roth spots:** Retinal hemorrhages with white centers.
- **Mycotic aneurysm:** Infection-induced ballooning of an artery.

Investigations

- **Blood culture:** Identifies specific microorganisms.
- **Echocardiography:** Imaging to detect vegetations or abscesses.
 - **Transoesophageal echocardiography:** A more detailed imaging technique.
- **ESR/CRP:** Markers of inflammation.
- **Haematuria:** Presence of blood in urine.

Duke's Criteria for Diagnosis

- **Major Criteria:**
 - **Positive blood culture:** Evidence of infection in the bloodstream.
 - **Regurgitation:** Backflow of blood due to valve dysfunction.
- **Minor Criteria:**

- **Embolic events:** Blockage in blood vessels caused by dislodged clots.
 - **Predisposing conditions:** Existing heart issues increasing infection risk.
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Principles of Management of Infective Endocarditis

Antibiotics

- **Bactericidal:** Kills bacteria directly.
- **Parenteral:** Delivered via injection or IV.
- **MIC (Minimum Inhibitory Concentration):** Lowest antibiotic concentration to inhibit bacteria.

Empirical Therapy

- **Subacute cases:** Ceftriaxone + gentamicin.
- **Prosthetic valve infections:** Add vancomycin.

Common Antibiotics

- **Penicillin G:** Targets gram-positive bacteria.
 - **Gentamicin:** Effective against gram-negative bacteria.
 - **Vancomycin:** Used for resistant infections.
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Surgical Indications

- **Prosthetic valve infection:** Infections involving artificial valves.
 - **Fungal endocarditis:** Infection caused by fungi.
 - **Heart abscess:** Pus-filled infection in heart tissue.
 - **Large vegetations:** Significant clumps of bacteria and platelets.
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Prophylaxis

- **Bacteraemia:** Presence of bacteria in the blood, especially after dental or surgical procedures.
- **Antibiotic regimen:**
 - **Amoxicillin:** A commonly used penicillin antibiotic.
 - **Clindamycin:** An alternative for penicillin-allergic patients.

Cyanosis

Definition

- **Reduced haemoglobin:** Form of hemoglobin unable to carry oxygen, causing bluish skin color.
- **Haemoglobin derivatives:** Abnormal forms of hemoglobin, such as methaemoglobin or sulphaemoglobin.
- **Capillary blood:** Blood within the smallest blood vessels where oxygen exchange occurs.

Common Areas

- **Malar eminences:** Cheekbone areas.

Types of Cyanosis

Central Cyanosis

- **Arterial oxygen saturation:** Percentage of oxygen bound to hemoglobin in arterial blood.
- **Haemoglobin derivatives:** Abnormal forms of hemoglobin.

Peripheral Cyanosis

- **Peripheral blood flow:** Circulation in limbs and extremities.
- **Vasoconstriction:** Narrowing of blood vessels reducing blood flow.

Causes of Central Cyanosis

Decreased Arterial Oxygen Saturation

- **Fallot's Tetralogy:** A congenital heart defect involving four abnormalities leading to mixed oxygenated and deoxygenated blood.
- **Eisenmenger's Syndrome:** Reversal of blood flow due to high pulmonary blood pressure from untreated congenital heart defects.
- **Congestive cardiac failure:** Inability of the heart to pump sufficient blood, leading to low oxygenation.
- **Interstitial lung disease:** Scarring in lung tissues impairing gas exchange.
- **Pulmonary arteriovenous fistula:** An abnormal connection bypassing the lungs' oxygenation process.

- **High altitude:** Low atmospheric oxygen levels reducing arterial oxygen content.
- **Cirrhosis of the liver:** Chronic liver damage that can cause blood vessel dilation and poor oxygenation.

Abnormal Haemoglobin Derivatives

- **Methaemoglobinaemia:** Hemoglobin that cannot effectively release oxygen.
 - **Sulphaemoglobinaemia:** Hemoglobin altered by sulfur, reducing its oxygen-carrying ability.
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Causes of Peripheral Cyanosis

Reduced Cardiac Output

- **Mitral stenosis:** Narrowing of the mitral valve, reducing blood flow to the body.
- **Shock:** Life-threatening low blood flow caused by circulatory failure.

Local Vasoconstriction

- **Peripheral vascular disease:** Blockage or narrowing of blood vessels in the arms or legs.

Syncope (Fainting)

Definition

- **Postural tone:** The body's ability to maintain an upright posture.
 - **Prodromal phase:** Early warning phase before fainting, showing mild symptoms.
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Mechanism of Syncope

- **Cerebral ischemia:** Inadequate blood flow to the brain, leading to oxygen deprivation.
 - **Vasodilatation:** Widening of blood vessels, reducing blood pressure.
 - **Cardiac output:** Amount of blood the heart pumps in a minute.
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Types of Syncope

1. Vasovagal Syncope

- **Vagus nerve:** A cranial nerve that controls heart rate and blood vessel dilation.
 - **Peripheral vascular resistance:** Resistance to blood flow in small arteries.
 - **Tachypnoea:** Abnormally rapid breathing.
 - **Bradycardia:** Abnormally slow heart rate.
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2. Cardiac Syncope

- **Complete heart block:** Total interruption of electrical signals in the heart.
 - **Stokes-Adams-Morgagni syndrome:** A sudden fainting episode caused by heart block.
 - **Paroxysmal tachycardias:** Sudden episodes of abnormally fast heart rates.
 - **Sick sinus syndrome:** Malfunction of the heart's natural pacemaker.
 - **Myocardial infarction:** Heart attack due to blocked blood flow to the heart muscle.
 - **Aortic stenosis:** Narrowing of the aortic valve in the heart.
 - **Hypertrophic cardiomyopathy:** Thickening of heart muscle, obstructing blood flow.
 - **Pulmonary embolism:** Blood clot in the lungs.
 - **Left atrial ball valve thrombus:** A movable blood clot in the heart's left atrium.
 - **Myxoma:** A benign tumor in the heart.
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3. Orthostatic Syncope (Postural Syncope)

- **Postural hypotension:** Sudden drop in blood pressure when standing.
 - **Vasoconstrictor reflexes:** Mechanisms to tighten blood vessels and maintain blood pressure.
 - **Idiopathic:** Occurring without a known cause.
 - **ACE inhibitors:** Medications used to lower blood pressure by relaxing blood vessels.
 - **Diabetic neuropathy:** Nerve damage caused by high blood sugar levels.
 - **Tabes dorsalis:** A late-stage complication of untreated syphilis, damaging nerves.
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4. Syncope Associated with Cerebrovascular Disease

- **Cerebrovascular:** Related to blood vessels supplying the brain.
 - **Occlusion:** Blockage in a blood vessel.
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5. Miscellaneous Causes of Syncope

- **Carotid sinus:** A pressure-sensitive area in the carotid artery that helps regulate blood pressure.
- **Glossopharyngeal neuralgia:** Severe nerve pain in the tongue and throat region.
- **Micturition syncope:** Fainting during or after urination.
- **Hypoglycaemia:** Low blood sugar levels.
- **Hyperventilation:** Breathing rapidly, causing low carbon dioxide levels in the blood.
- **Hysterical fainting:** Psychologically induced fainting episodes.

METRONIDAZOLE

Dosage:

- **t.d.s.:** Thrice a day.
- **Extra-intestinal:** Infections outside the intestines.

Mechanism of Action:

- **Nitroimidazole:** A class of antibiotics that contains a nitro group and an imidazole ring, which is effective against anaerobic bacteria and protozoa.
- **Electron transport proteins:** Proteins in the microorganism that transfer electrons during metabolic processes.
- **Nitro group:** A chemical group ($-\text{NO}_2$) that is activated when it is reduced inside the cell to cause damage.
- **DNA strand breakage:** The breaking of the DNA molecule, which impairs the ability of the cell to replicate or repair itself.
- **Anaerobic bacteria:** Bacteria that do not require oxygen to survive and grow.

Indications:

- **Giardiasis:** An infection caused by the protozoan **Giardia lamblia**, affecting the gastrointestinal tract.
- **Trichomonas vaginitis:** An infection of the vagina caused by the protozoan **Trichomonas vaginalis**.
- **Dracunculosis:** An infection caused by the **Guinea worm** (*Dracunculus medinensis*), resulting from drinking contaminated water.
- **Ulcerative gingivitis:** A gum infection characterized by ulceration and inflammation, often caused by bacteria.
- **Amoebiasis:** An infection caused by **Entamoeba histolytica**, a protozoan, affecting the intestines and sometimes the liver.

- **Anaerobic infections:** Infections caused by bacteria that thrive in oxygen-free environments, often affecting wounds or abscesses.

Side Effects:

1. **Gastrointestinal (GI):**
 - **Nausea:** The feeling of wanting to vomit.
 - **Vomiting:** The act of throwing up.
 - **Diarrhea:** Frequent, loose, or watery stools.
 - **Metallic taste:** A strange, metallic flavor in the mouth.
 - **Abdominal pain:** Pain in the stomach or intestines.
2. **Central Nervous System (CNS):**
 - **Headache:** Pain or discomfort in the head.
 - **Dizziness:** A sensation of lightheadedness or unsteadiness.
 - **Vertigo:** A spinning or whirling sensation, often leading to loss of balance.
 - **Ataxia:** A lack of muscle coordination, affecting movement and balance.
3. **Skin and Allergic Reactions:**
 - **Urticaria:** Red, itchy hives on the skin.
 - **Pruritus:** Itching.
 - **Flushing:** Redness of the skin, often on the face, caused by increased blood flow.
4. **Antabuse-like Reaction:**
 - **Acetaldehyde dehydrogenase:** An enzyme that helps break down alcohol in the body.
 - **Acetaldehyde:** A toxic compound that builds up when alcohol is consumed, leading to symptoms like nausea and vomiting.

Important Notes:

- **Liver disease:** Conditions that affect the function of the liver, such as hepatitis or cirrhosis.
- **Contraindicated:** Not recommended or safe to use in certain conditions, such as during the first trimester of pregnancy.
- **Alcohol:** Should be avoided during treatment due to potential interaction with the drug leading to unpleasant reactions.

CPR

- **Ventilation:** Process of supplying oxygen to the lungs.
- **Perfusion:** Flow of blood to tissues and organs.
- **Spontaneous:** Occurring naturally without external intervention.
- **Cardiopulmonary:** Relating to the heart and lungs.

Phases of Life Support

- **Prolonged:** Continuing for an extended period.
- **Post-resuscitation:** Care given after revival from cardiac arrest.

Change in CPR Sequence

- **Defibrillation:** Delivery of an electric shock to restore a normal heart rhythm.

Circulation

- **Xiphoid process:** Small cartilage at the lower end of the sternum.
- **Pulsations:** Rhythmic throbbing of the arteries.
- **Reassessment:** Rechecking or re-evaluating.

Airway Management

- **Obstruction:** Blockage preventing normal flow.
- **Mandible:** Lower jawbone.
- **Subdiaphragmatic:** Below the diaphragm.
- **Expels:** Forces something out.

Breathing

- **Seal:** Close securely.
- **Visible chest rise:** Chest movement indicating air entering the lungs.
- **Tidal volume:** Amount of air moved in one breath.

ALS Techniques

- **Intubation:** Insertion of a tube into the airway.
- **Cricothyrotomy:** Emergency procedure to open the airway at the neck.
- **Tracheostomy:** Surgical opening in the trachea to aid breathing.
- **Ventilator:** Machine that supports or controls breathing.

Defibrillation

- **Ventricular tachycardia:** Abnormal rapid heart rhythm starting in the ventricles.
- **Biphasic/Monophasic:** Types of defibrillator waveforms.

Drugs in ALS

- **Amiodarone:** Drug used to treat irregular heart rhythms.
- **Torsade de Pointes:** Specific type of abnormal heart rhythm.
- **Digoxin toxicity:** Overdose or adverse reaction to the drug digoxin.
- **Hyperkalemia:** High levels of potassium in the blood.

Post-Resuscitation Care

- **Vasoactive drugs:** Medications that affect blood vessel tone and heart function.
- **Hyperglycemia:** High blood sugar levels.
- **Neurological outcome:** Brain function and recovery after an event like cardiac arrest.